Memory-Contrastive Unsupervised Domain Adaptation for Building Extraction of High-Resolution Remote Sensing Imagery

Jie Chen, *Member, IEEE*, Peien He, Jingru Zhu, Ya Guo, Geng Sun, Min Deng, and Haifeng Li, *Member, IEEE*

***Abstract*— Deep learning-based semantic segmentation has been widely applied for building extraction. However, due to the domain gap, the extraction of building in high-resolution remote sensing imagery is difficult when the model trained on a source dataset is directly used to test on a target data. Considering that humans can retrieve memory to deal with correlative tasks in different domains, memory mechanisms have been developed effectively to assist cross-domain feature extraction. Besides, the effectiveness of memory mechanisms highly depends on the correlation of memory to the task. Therefore, the domain-invariant memory is crucial in cross-domain building extraction task. To this end, a memory-contrastive unsupervised domain adaptation (DA) method is proposed on the basis of a novel memory mechanism. Specifically, to facilitate the model to memorize domain-invariant features, we first conduct a normalization-based image style transfer strategy and a discriminator-based adversarial method at the image level and feature level, respectively. Subsequently, we carry out a memory-contrastive module to obtain domain-invariant features. Especially, a teacher–student network is exploited to help knowledge transferring by knowledge distillation to enhance the performance of the memory-contracted (MCD) module. To narrow the distance between the two domains, a memory bank is designed to store and update category features obtained from the source domain, and then, the similarity between category features in the target domain and memory bank is calculated. Results of the cross-domain experiments show that the proposed method can achieve optimal building extraction (github: https://github.com/RS-CSU/MDANet).**

***Index Terms*— Building extraction, contrastive learning, domain adaption, memory, semantic segmentation.**

I. Introduction

With the development of remote sensing (RS) technology, the data with different spatial resolutions, multiple spectral characteristics, and multimode become increasingly available. The investigation of buildings from RS images is significative for human exploration and understanding of geographical and ecological environment, especially in urban construction and planning [1], [2], [3], natural disaster and crisis management [4], [5], and other applications. Previous studies on building extraction focused on features design and extraction, such as color [6], spectrum [7], edge [8], shape [9], texture [10], and shadow [6]. However, these artificially designed features are prone to bias under different lighting, atmospheric conditions, sensor quality, scale, and environment because of the complexity of building structures and the strong similarity with other classes, such as roads. This requires the related methods to only solve specific building extraction task with specific data source.

In recent years, with the superiority of graphic processing unit (GPU) and the availability of a large number of training datasets, deep learning has been widely investigated. With the advent of fully convolutional network (FCN) [10], FCN-based deep learning algorithms have been gradually used in the field of computer vision, such as in medical image analysis [11], autonomous driving [12], and RS image interpretation [13]. Especially in RS field, many applications use deep learning means, such as object detection [14], cloud detection [15], scene classification [16], change detection [17], and land cover classification [18], to achieve their goals. Meanwhile, many research on deep learning-based building extraction [19], [20], [21] have been proposed. Basically, these works require a large amount of labeled data to support model training. In fact, due to the differences in sensors and imaging conditions, the domain differences between source and target datasets can cause the model to perform poorly and generalize insufficiently. In general, the source domain represents a domain different from the test sample but has rich supervision information, while the target domain represents the domain where the test sample is located with no labels or only a few labels. Source and target domains often belong to the same class of tasks, but are distributed differently. Thus, overcoming this domain gap and realizing effective knowledge transfer among domain are still open issues that need to be solved.

As the domain discrepancy between source and target datasets prevents the model from producing satisfactory results, unsupervised domain adaptation (UDA) has become a main strategy for reducing the domain gap, thus realizing the transfer of knowledge. At present, many UDA methods have been developed to realize building recognition. Shi et al. [22] proposed a UDA method by aligning the pixel-level and representation-level networks, respectively. The pixel-level domain adaptation (DA) transforms the source domain data

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The authors are with the School of Geosciences and Info-Physics, Central South University, Changsha 410083, China (e-mail: cj2011@csu.edu.cn; lihaifeng@csu.edu.cn).

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into the target domain data through the cycle generative adversarial network (CycleGAN). The representation-level domain adaptation can be achieved by implementing adversarial learning between source and target domain images. This kind of method took a generative adversarial network (GAN)-based module to transfer the knowledge of the images, so that the source domain image could be aligned with the target domain image. However, to train a GAN-based network with stable output, training procedures with complex loss functions need to be optimized. Especially for high-resolution remote sensing imagery (HRSI) with complex backgrounds, achieving satisfactory results using the style transfer-based methods is often difficult. Therefore, many researchers transfer the key knowledge from the feature level to achieve the purpose of domain adaptation. For example, Li et al. [23] proposed multiple weakly supervised constraints to reduce the adverse effects of data shift. However, the target domain data had not been fully utilized for these methods, thus resulting in the insufficient transferability of features.

In the UDA building extraction task, because the target domain data have no labels, the information of the image itself must be explored fully to obtain reliable features. Nowadays, the self-supervised learning model can be pretrained using a large number of unlabeled images and then be fine-tuned on a downstream task with a small number of labeled samples [24]. For example, global style and local matching contrast learning network (GLCNet) [25] used a global style contrastive learning module to learn image-level representations and designed local features to match the contrastive learning module to learn local region representations. Zhao et al. [26] developed a new specificity enhancement module to generate sample pairs and then leveraged a contrastive learning model to extract features from the sample pairs. The contrastive learning method can extract essential features to assist the downstream tasks. Consequently, this article also conducts contrastive learning on the target domain to obtain domain-invariant features through similarity measurement.

Humans can perceive and understand something new based on their previous memory. Human semantic memory [27] can be extracted from previous experiences in a reusable form and generalized into different cognitive activities [28], [29]. The domain-invariant building features should be relevant memory because the features of the same class, even when obtained from different datasets, have commonalities [32]. Based on this analysis, a memory-contracted domain adaptation network (MDANet) is proposed for building extraction by utilizing the information of labeled source data and enhancing the representation of key features of the target domain. First, to make the variance and mean of the source data similar to the untrained target data, the image is normalized from the s by using Wallis filter (WF). Second, the constraints are imposed at the feature level, and the adversarial strategy is introduced to reduce the feature differences between source data and target data. Subsequently, a memory-contracted (MCD) module is proposed on the basis of the positive-pair-only contrastive learning method [30] and human memory mechanism to further reduce the semantic gap. Specifically, to make the MCD module work well, a teacher–student mode is applied to help transfer knowledge. In addition, to fully utilize the key features from the whole source dataset, a memory bank is exploited to store and reuse the class features extracted by the teacher network from the source data. Subsequently, the category features from the target data are extracted by the student network, and optimization is performed by calculating contrastive loss with the features in the memory bank. In addition, the pseudo-label is made for target data to support the self-training. In brief, our main contributions are presented as follows.

1) A human memory mechanism-inspired UDA network called MDANet is proposed for cross-domain building extraction.
2) A positive-pair-only contrastive learning-based MCD module is developed to extract domain-invariant building features.
3) A memory bank embedded MCD module is designed to reduce the semantic gap between source and target data for the domain-invariant representation.

The remainder of this article is organized as follows. Section II describes the related work. Section III illustrates our proposed method. Section IV presents the experimental results, and Section V concludes this work.

II. Related Works

This section illustrates the related deep learning-based building extraction, including the following four parts: full-supervised building extraction, UDA in RS field, memory bank, and contrastive learning.

A. Full-Supervised Deep Learning-Based Building Extraction

Given that building extraction is a binary segmentation task in RS field, many FCN-based methods [31], [32], [33] have been applied for this task with the development of FCN [10]. With further research on various technologies, including but not limited to attention mechanism, co-occurrence visual mode, and multiscale fusion strategy, more and more deep learning-based semantic segmentation methods, such as UNet [34], Deeplab series [35], [36], [37], [38], pyramid scene parsing network (PSPNet) [39], and squeeze-and-excitation network (SENet) [40], are proposed and promoted the technological progress in building extraction. For example, dense boundary block [41] has been designed on the basis of dense block, boundary block, and spatial channel attention block to solve the problem of blurred boundaries and intraclass inconsistency. A dense residual neural network [42] was proposed to reduce the amount of network parameters without any loss of accuracy. Guo et al. [43] categorized the input images by multilabeled scenes and mapped the buildings in different scenes at the pixel level. However, these methods require a large amount of labeled data to train the model. Also, due to the existence of domain shift, obtaining satisfactory results is often difficult when using a model trained on one dataset but tested on another dataset.

B. UDA Semantic Segmentation

At present, UDA technology is also widely used in the field of semantic segmentation of RS images. For example,

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Chen et al. [44] used feature alignment to narrow the semantic-level differences between source and target domains on the one hand and utilized a category deterministic attention module to enhance attention to category-level unaligned features on the other hand. Full-level domain adaptation network (FDANet) [45] is a novel full-layer domain adaptive network that combined image, feature, and output-level information efficiently to extract buildings based on a style transfer operation, a mean-teacher model, and a self-training strategy. Mei et al. [46] proposed a small-sample scene classification framework based on deep feature fusion, which extracted multilevel features from different layers of the CNN without labeling samples. Zhang et al. [47] proposed subspace correlation maximization and dynamic statistical distribution alignment to balance the influence of source and target domains for RS image scene classification. A DA method was proposed to fuse synthetic aperture radar (SAR) and optical features to use their complementary information while aligning multitemporal by reducing spectral and radiometric difference to improve the change detection accuracy [48]. The correlated alignment DA and adversarial DA were applied into the region-based vehicle detector to improve detection accuracy in the target domain [49]. Cai et al. [50] optimized the segmentation model from two opposite directions: in the direction of source→target, the source domain images are transformed into images with the style of the target domain, and the classifier is adapted to the target domain. In the opposite direction, the target domain images are transformed into images with the style of the source domain, and the source classifier is optimized. Benjdira et al. [51] designed a two-step algorithm: a GAN model is first used to transform the source domain into the target domain style, and then, the mapped labels were used to fine-tune the already trained semantic model. In general, DA task can be completed in two ways: one is by using an image-level alignment method, such as the GAN-based style transfer model, and another one is by aligning the source domain with the target domain in the feature space. However, adapting the HRSI by only applying image-level alignment is difficult, and the extracted features are not always essential to achieve a good transfer of knowledge.

C. Memory Bank

Memory bank can introduce auxiliary memory units to store and reuse the task-related information to enhance the network capacity. Neural turing machines [52] enhanced the memory capacity by using the external memory module where the controller writes information to the memory module via write heads and reads information from the memory module using read heads. Weston et al. [53] used a memory component to store scene information to achieve the function of long-term memory for the first time. Sukhbaatar et al. [54] proposed an end-to-end model that could be used as a memory network in a continuous form. References [55], [56], and [57] used external memory unit in semisupervised semantic segmentation and DA. In the domain adaptation task, the source domain is labeled, and the information obtained from the source domain is relatively reliable. This reliable knowledge can be stored by using the memory bank and can be reused when needed to assist in knowledge transfer.

D. Contrastive Learning

Contrastive learning, as an important member of self-supervised learning, has been widely used in computer vision, natural language processing, and other fields. It aims to bring the augmented samples as close to each other as possible in the embedding space and then keep the different samples as far away from each other as possible. Contrastive method has achieved great success in computer vision tasks through unsupervised contrastive pretraining, especially when the number of negative samples is large because more negative samples can cover the underlying data distribution more likely. However, the number of negative samples is always limited by the batch size. A momentum contrast is recently proposed to solve this problem by maintaining a large number of negative sample queues [58]. In particular, this method does not use backpropagation to update the negative encoder but uses momentum to update the negative encoder periodically. Simple framework for contrastive learning of visual representations (SimCLR) [59] introduced nonlinear transformations and adversarial loss functions to improve the quality of learnable representation. Instead of using negative samples, bootstrap your own latent (BYOL) [60] applied the stop-gradient operation to an encoder and updated it using momentum method. In addition to not using negative sample pairs, Simsiam [30] proposed a novel "stop-gradient" strategy so that it learned meaningful representation without using large batch and momentum encoders. Furthermore, Wang et al. [55] exploited unreliable pseudo-labels as a supplement to labeled samples to improve model performance with contrastive learning framework. Huang et al. [57] proposed a new category contrast technique that introduced semantic priors based on instance recognition for visual UDA tasks. In this work, a memory bank is designed to store reliable building features obtained from the source domain and used the features by utilizing contrastive learning. The constraint of contrastive loss is imposed on the distance shortening between the source domain and target domain, and the features in the memory bank would become more domain-invariant.

III. Methodology

A. MDANet Architecture

The MDANet framework consists of a WF, teacher–student model, discriminator, and MCD module, which is illustrated in Fig. 1. We define source data as $\mathcal{D}_s$, which consists of image x_s and label y_s, and target data as $\mathcal{D}_t$, which includes image x_t. To reduce the semantic gap caused by the spectral difference between the source image x_s and the target image x_t and thus making it easier for the model to extract domain-invariant features for memorization, we use a WF to transform x_s into x_t. Subsequently, the transformed source image and the target image are inputted into the student network G_{stu}. Then, at the end of the encoder in G_{stu}, for the preliminary alignment in feature space to enforce the model to extract domain-invariant features, a discriminator $\mathcal{D}_f$ is exploited to identify whether

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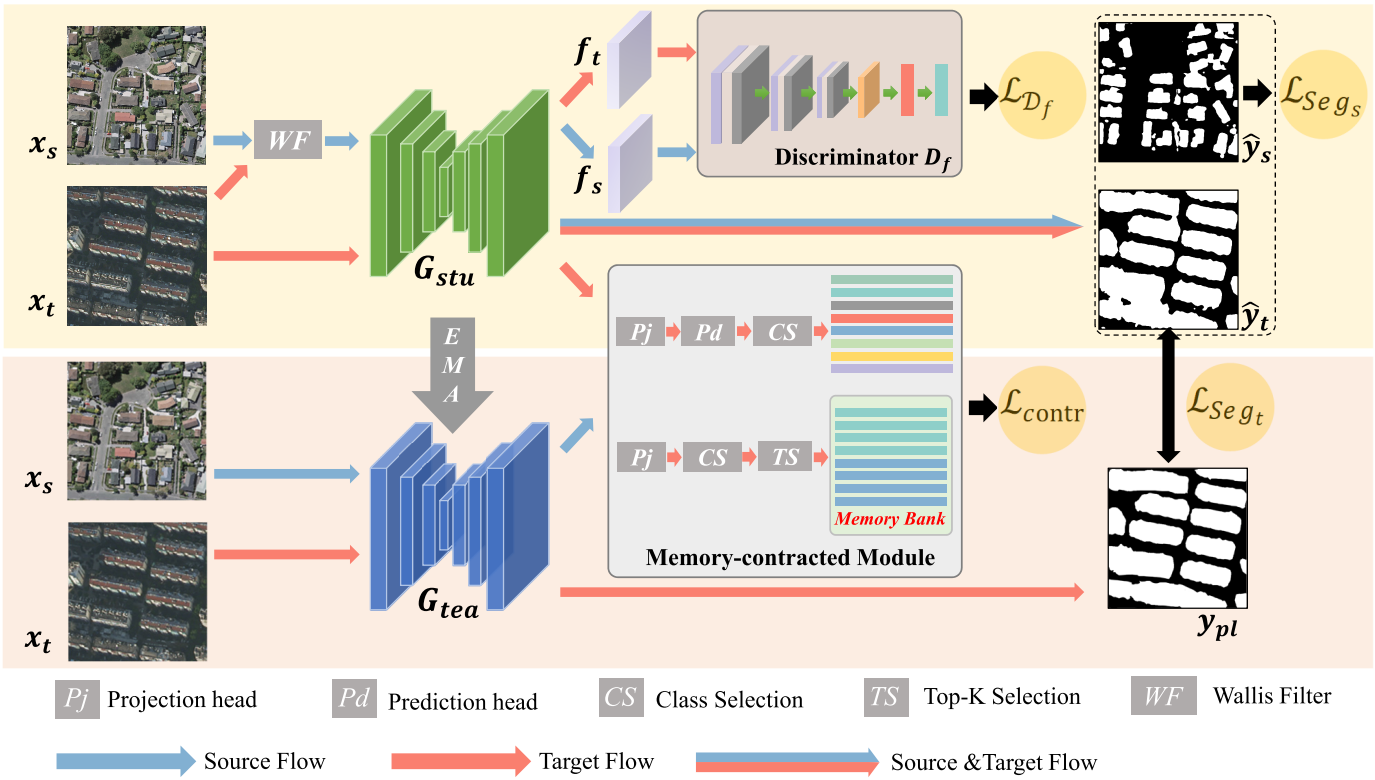

Fig. 1. Overview of MDANet.

features f_s and f_t are from the same or different domains. The result can be used to calculate the adversarial loss $\mathcal{L}_{\mathcal{D}_f}$. Meanwhile, after the end of the decoder, the prediction results of source image and target image $\hat{y}_s$ and $\hat{y}_t$ are generated. The segmentation loss of the source data $\mathcal{L}_{Seg_s}$ can be obtained by calculating the cross entropy of label y_s and segmentation result $\hat{y}_s$. For the target segmentation map, we use a teacher network G_{tea} that is updated by exponential moving average (EMA) to generate corresponding pseudo-label y_{pl} to obtain the segmentation loss of target data $\mathcal{L}_{Seg_t}$. As a result, the knowledge of the student network is transferred to the G_{tea} through EMA, and G_{stu} can learn more source knowledge based on the pseudo-label [61]. For the target domain data, we input the feature map obtained from the student network into the projection and prediction heads, which consist of Linear, BatchNorm, and rectified linear unit (ReLU), and select categories according to the pseudo-labels to obtain multiple features vectors of each category of the target domain. To utilize the key knowledge that appeared in the source data, a memory bank is designed. Through the transfer of knowledge from G_{stu} to G_{tea}, G_{tea} can extract features. Moreover, G_{tea} can benefit from longer term memory, which can push the model to learn smoother output [61]. Therefore, the memory bank can be established by feeding the source image x_s into the teacher network G_{tea}. The projected feature map can be obtained by projection transformation. Then, the feature vectors that satisfy the confidence threshold are selected from the projected feature map according to category based on the predicted probability of the source domain obtained by the teacher network. Subsequently, top-k vectors are selected according to the confidence and are stored in the memory bank. The contractive loss $\mathcal{L}_{contr}$ is used to penalize the inconsistency between the source image features from the memory bank and the target image features. Through the above steps, the key knowledge from the labeled source data can be transferred effectively, thus enabling the student network to take effect on unlabeled target data.

B. Wallis Filter in Image-Level Adaption

Although the image style transfer-based domain adaption methods have made tremendous success, they cannot be directly implemented on RS images because training a stable GAN-based model for information-rich and scene-complex HRSIs is difficult. Inspired by [62], the color consistency method, WF [63] which is widely used in RS field, is exploited to replace GAN-based model. It realizes the color transfer of the images by mapping the mean values $\mu(\cdot)$ and standard deviations $\sigma(\cdot)$ of images linearly. Specifically, a batch of images of the same number is randomly selected in the target domain and the source domain, respectively. In each iteration, the corresponding $\mu(\cdot)$ and $\sigma(\cdot)$ are calculated using the data of the current batch. To further illustrate the process, it assumes there are three images S1, S2, and S3 in a batch of source domain, and T1, T2, and T3 are the images in a batch of target domain. The mean $\mu(x_t)$ and standard deviation $\sigma(x_t)$ of T1, T2, and T3 are calculated to represent their image style, and the mean $\mu(x_s)$ and standard deviation $\sigma(x_s)$ of S1, S2, and S3 are calculated similarly. Subsequently, the source image x_s can be transferred into the image x_{st} similar to the target domain according to the following formula:

$$x_{st} = \lambda \left[\sigma(x_t) \left(\frac{x_s - \mu(x_s)}{\sigma(x_s)} \right) + \mu(x_t) \right] + (1 - \lambda) x_s. \quad (1)$$

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C. Discriminator in Feature-Level Adaption

After the WF is used to reduce the domain gap between source and target images in the input level, a discriminator is exploited to reduce difference in feature level further. The smaller the domain difference of the features of the same category in different domains is, the more likely the discriminator judge that they come from the same domain. In particular, as shown in Fig. 2, three continuous (Conv)–(Leaky-ReLU)–(Dropout) modules are used to extract condensed high-dimensional semantic features. Furthermore, through the global average pooling (GAP), fully connected (FC), and Sigmoid layers, the discriminator gains the result value as 0 or 1, where the discriminator would output 1 if it determines that two features come from the same domain.

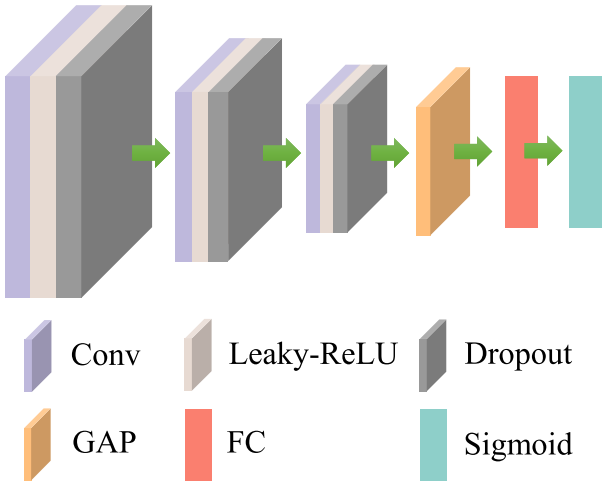

Fig. 2. Architecture of discriminator.

D. Teacher–Student Model

Teacher–student model can realize the information interaction between two subnetworks through a knowledge distillation strategy. Teacher–student model consists of a teacher model G_{tea} and a student model G_{stu} with the same structure. It transfers the knowledge from the teacher model to the student model while exploiting the parameters of the student model to update the parameters of the teacher model [62], [63], [64]. In this article, through EMA, the parameters of G_{stu} are used to update G_{tea} in each iteration, and G_{tea} generates corresponding pseudo-labels to supervise the prediction results of the student network on the target domain. Fig. 1 shows that the target domain prediction results $\hat{y}_{\text{t}}$ generated by the student network G_{stu} calculate the loss with the pseudo-label y_{pl} obtained by the teacher network G_{tea}, and the memory bank composed of the feature vectors extracted from the source domain is compared with the features extracted from the target and the student network G_{stu}. In the whole process, the teacher model is updated by using the EMA so that the key knowledge in the student can be transferred. Specifically, during the model training stage, only the student network updates the model parameters with gradient descent, whereas the teacher network updates according to the way that the parameters of student network uses moving average. The update means of the teacher network can be expressed by the following formula:

$$\omega_i^{\text{tea}} = \alpha\omega_{i-1}^{\text{tea}} + (1-\alpha)\omega_i^{\text{stu}} \tag{2}$$

where ω is the parameter of the model, and i and α represent the ith iteration and the degree how the updated teacher network relies on the student network parameters in the current iteration, respectively.

E. Memory-Contracted Module

Regardless of which domain the image comes from, the same category should have domain-invariant features. This domain-invariant feature is the key knowledge that needs to be transferred. Therefore, inspired by positive-pair-only contrastive learning [30] and human memory mechanism, we propose the MCD module, as presented in Fig. 1. To fully utilize the information of the labeled source domain, a memory bank is designed to store and reuse the source feature, which is extracted by the teacher model G_{tea} with high confidence. Meanwhile, the student model G_{stu} extracts the feature vectors from the target data. In particular, we extract the feature vectors for each category, and each vector can be considered a form of feature for each category. Considering that the memory bank contains high-quality classwise features learned from labeled samples of the source domain, the contrastive loss can help obtain better interclass variance in the feature space and align the distribution of unlabeled data of the target domain. The features stored in the memory bank are continuously updated with the model training, and the stored features possess domain-invariant characteristics.

In detail, let G^- be the network without classification layer, and x, y be the source data $x_{\text{s}}, y_{\text{s}}$ or target data with pseudo-label $x_{\text{t}}, y_{\text{pl}}$, respectively. For the target data input to G_{stu}^-, the feature vector V can be extracted by $G_{\text{stu}}^-(x_{\text{t}})$. Then, V will be fed into the projection head P_j to obtain Z by $P_j(V)$. Projection head is a common construction in contrastive learning. It is a simple multilayer perceptron (MLP) with the structure: Linear–Batchnormal–ReLu–Linear. It aims to change the embedding space of features, mapping the representation to the space of contrastive loss. Then, the prediction head P_{d} can take Z as its input, that is, $P = P_{\text{d}}(Z)$. Subsequently, every class's vector $P_{\text{c}} = p_{\text{c}}$ can be extracted by y_{pl}. In detail, y_{pl} can be resized to the same size as the feature which is before inputting to the classifier, and the building's features on the feature map can be extracted according to the position of the building pixel on the pseudo-label. For the source data input for G_{tea}^-, the method of obtaining Z' is consistent with that of Z.

The memory bank is an external memory module that consists of many feature vectors. To obtain the reliable and class-related feature vectors stored in the memory bank, the teacher network is used to predict the input source domain image to obtain a probability map, and then, $Z'_{\text{c}} = z'_{\text{c}}$ that corresponds to each category can be obtained according to the set confidence threshold value (details in Section IV-B). Using this value and ranking every probability value in the probability map, we update the memory bank by only using

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the vectors of the top-k. Due to the limited capacity of the memory bank, the "First in First out" rule is adopted in this article. In the process of iterative updating of the model, the old memory is constantly overwritten with new memory.

Next, to utilize these feature vectors in contrastive way, the relationship between p_c and z'_c must be explored. The contrastive loss encourages the intraclass variance of the prediction vectors p_c and the class vectors z'_c stored in memory bank to decrease. For this purpose, the cosine similarity is used to be the similarity measure S

$$S(p_c, z'_c) = \frac{p_c \cdot z'_c}{\|p_c\| \cdot \|z'_c\|}. \tag{3}$$

F. Loss Functions

Suppose the source domain dataset is represented as $\mathcal{D}_s = \{x_s, y_s\}$ and the target domain dataset is represented as $\mathcal{D}_t = \{x_t\}$.

1) Segmentation Loss: On the one hand, buildings should be extracted from the source domain data to reflect that the model can capture building features accurately. On the other hand, buildings also need to be extracted accurately on the target domain to indicate that the knowledge has been transferred successfully. Therefore, two binary cross-entropy losses can be used to constrain the model. For source data, $\mathcal{L}_{\text{Seg}_s}$ can be defined as follows:

$$\mathcal{L}_{\text{Seg}_s} = -\frac{1}{N_s}\sum_{i=1}^{N_s} y_s \cdot \log(\hat{y}_s) + (1 - y_s) \cdot \log(1 - \hat{y}_s) \tag{4}$$

where $\hat{y}_s$ is the prediction result, and N_s denotes the batch size of the source images.

Since the labels of the target domain images are unknown during training, the prediction results of the target domain data are supervised using the online generated pseudo-labels. The strategy that retraining the model based on pseudo-labels has been widely used in the UDA field to improve the generalization ability of the model for the target domain. However, the quality of pseudo-labels is uneven. To prevent inaccurate pseudo-labels from affecting the model negatively, the pseudo-labels are filtered according to confidence, and the loss function is weighted with the probability values that correspond to the selected pseudo-labels

$$\mathcal{L}_{\text{Seg}_t} = -\text{O}(p(y_{\text{pl}})) \cdot \frac{1}{N_t}\sum_{i=1}^{N_t} y_{\text{pl}} \cdot \log(\hat{y}_t) + (1 - y_{\text{pl}}) \cdot \log(1 - \hat{y}_t) \tag{5}$$

$$\text{O}(p(y_{\text{pl}})) = \begin{cases} p(y_{\text{pl}}), & p(y_{\text{pl}}) \geq \text{confi.value} \\ 0, & x < \text{confi.value} \end{cases} \tag{6}$$

where y_{pl}, $\hat{y}_t$, and $p(\cdot)$ denote the pseudo-labels, prediction result, and probability map, respectively.

2) Adversarial Loss: To help model mining reliable domain-invariant feature after conducting WF in the input level, the adversarial loss based on discriminator $\mathcal{D}_f$ is set up.

The discriminator $\mathcal{D}_f$ can determine whether the input feature map is from the source domain or the target domain. For the discriminator itself, the source of the feature map must be determined accurately. For the network that generates the feature map, the target domain image must be used to generate the source domain-like features that can deceive the discriminator. Therefore, this loss can be divided into two parts

$$\mathcal{L}_{\mathcal{D}_f} = \frac{1}{N_s}\sum \text{BCE}(f_s, 1) + \frac{1}{N_t}\sum \text{BCE}(f_t, 0) \tag{7}$$

$$\mathcal{L}_{\text{adv}} = \frac{1}{N_t}\sum \text{BCE}(f_t, 1) \tag{8}$$

where f is the feature map, and BCE denotes the binary cross-entropy.

3) Memory Contrastive Loss: As mentioned above, in the training stage, the memory bank generated by the teacher model using source data consists of reliable category feature vectors z'_c whereas the feature vectors p_c extracted by the student model may have different distributions with z'_c in memory bank, which means that the features of the same category in the source and target domains still have semantic gap. As such, the memory contrastive loss $\mathcal{L}_{\text{contr}}$ is designed to align the source data distribution with the target data distribution. Equation (3) calculates the similarity S between z'_c and p_c. Thus, memory contrastive loss can be defined as

$$\mathcal{L}_{\text{contr}} = \frac{1}{N_{p_c}}\frac{1}{N_{z'_c}}\sum_{p_c \in P_c}\sum_{z'_c \in Z'_c}(1 - S(p_c, z'_c)) \tag{9}$$

where N_{p_c} and $N_{z'_c}$ indicate the number of vectors. The number of target domain features obtained from the student network is not limited, and the similarity between these features ($N_{p_c} * C$, C denotes the feature channel) and the features in the memory bank ($N_{z'_c} * C$) is calculated ($N_{p_c} * N_{z'_c}$, in the form of a matrix).

Based on the above details, the final loss can be defined by combining the previous components

$$\mathcal{L} = \lambda_{\text{Seg}_s}\mathcal{L}_{\text{Seg}_s} + \lambda_{\text{Seg}_t}\mathcal{L}_{\text{Seg}_t} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} + \lambda_{\text{contr}}\mathcal{L}_{\text{contr}} \tag{10}$$

where λ is the weight of the corresponding loss function.

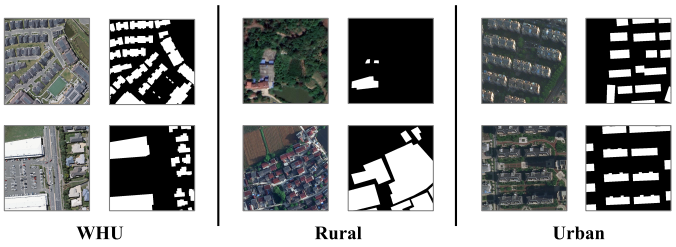

Fig. 3. Example images of datasets.

IV. Experimental Results and Discussion

A. Datasets

To verify the effectiveness of the method, two distinctive RS datasets, namely, LoveDA [67] and WuHan University (WHU) building dataset [68], are used in our experiments, where LoveDA contains buildings in two different scenes. Table I provides an attribute comparison of the three building extraction datasets, and Fig. 3 shows some data.

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TABLE I
DATASETS FACT SHEET

Dataset	Resolution(m)	Size(pixel)	Tiles	Area
LoveDA				
-Urban	0.3	512 × 512	5616	China
-Rural	0.3	512 × 512	3230	China
WHU	0.3	512 × 512	8189	New Zealand

1) LoveDA: LoveDA is a high spatial resolution land cover dataset for domain adaptive semantic segmentation [65]. It has the characteristics of diverse object scales, complex scenes, and inconsistent feature distribution. The LoveDA dataset consists of 0.3-m images with a size of 1024 × 1024 pixels obtained from many cities in China, such as Nanjing and Wuhan, covering 536.15 km^2, which includes urban and rural scenes. The urban scene is more man-made objects, and their styles and scales are richly diverse. Meanwhile, for the rural scene, natural objects are the main land cover class with man-made objects, such as buildings, belong to the long-tail class. This dissimilarity gives rise to the difference in the distribution of rural and urban data, which makes the transfer of knowledge between these two scenes to achieve the requirements of domain adaptation tasks difficult. For extracting building, only the label of the building is required, and other classes' label can be regarded as background. Since other classes, except buildings, are set as background classes, the proportion of negative classes represented by background classes is high. Capturing building features is not beneficial for the model, so the data that belong to all background classes are eliminated. All images are cropped to 512 × 512 pixels. After processing, 5616 images are in the Urban dataset and 3230 are in the Rural dataset, of which 80% of the data are used as training set and the rest are used as test set, respectively.

2) WHU Building Dataset: The dataset is based on the aerial imagery that contains 220 000 individual buildings at a resolution of 0.075 m and 450 km^2 covering in Christchurch, New Zealand. It includes buildings in different scenes, such as countryside, residential, culture, and industrial areas. Extensive experiments in the past have demonstrated that the performance of the FCN method cannot be improved significantly when the resolution is larger than 0.3 m. Therefore, 0.075-m resolution aerial imagery is downsampled to 0.3-m resolution image, and all images are cropped into 8189 small image patches of 512 × 512 pixels, where 80% of the images are used to build the training set.

B. Implementation Details

For the basic framework of G_{stu} and G_{tea}, DeepLabV3+ [38] with a ResNet101 [66] backbone is adopted, which is similar to many previous works [45], [67], [68]. In the training stage, stochastic gradient descent (SGD) with a momentum of 0.9 is used as optimizer for G_{stu}, and the base learning rate is set to 4e−4 with a weight decay of 5e−1 whereas Adam with betas 0.5 and 0.999 is exploited to update the parameters of $\mathcal{D}_{\text{f}}$. To demonstrate that the proposed method is effective in multiple differentiated scenarios, experiments are carried out on the datasets mentioned previously, and the bidirectional implementation is carried out by exchanging the source domain and the target domain, including Rural⟶Urban, Urban⟶Rural, Rural⟶WHU, WHU⟶Rural, Urban⟶WHU, and WHU⟶Urban. The batch size is set as 4 for the method, and the input image size is the original size of all the datasets, as shown in Table I. To maximize the transformation of the source domain image to the target domain style image at the input level, λ in (1) is set as 1. For the confidence value of constructing memory bank (Section III-D), it is set as 0.85 to filter inaccuracy and false feature that prevent the memorization of uncorrelated knowledge. To realize the knowledge transfer between teacher network G_{tea} and student network G_{stu}, α in (3) is set as 0.99 to implement knowledge transferring from G_{stu} to G_{tea}. To take advantage of correctly classified pseudo-label regions and minimize the influence of untrusted regions, the confidence is set to 0.8, that is, when the probability value is less than 0.8, the weight value is 0; otherwise, the weight value is equal to the probability value [see (5) and (6)]. The weight of the loss function is empirically set as follows: $\lambda_{\text{Seg}_s} = 1$, $\lambda_{\text{Seg}_t} = 1$, $\lambda_{\text{adv}} = 0.001$, and $\lambda_{\text{contr}} = 0.1$. To ensure that the model can extract category features and obtain reliable vectors stored in memory bank and pseudo-labels accurately, λ_{adv} and λ_{contr} are set as 0 in the first 1k iterations. The size of the memory bank is set as 256. Given that only two classes, namely, building class and background class, are in this building extraction task, only the feature vectors of the buildings are stored. For each image, the number of stored feature vectors is set as min(eligible, 64) according to the rank of probability. A NVIDIA GTX 1080Ti GPU with 11 GB is used for the proposed method, and Pytorch framework is used to implement it.

C. Comparative Methods

In this section, several state-of-the-art (SOTA) UDA methods are compared with the proposed method.

1) AdaptSegNet: AdaptSegNet [69] is an adversarial learning method for domain adaptation in semantic segmentation task. Considering that the semantic segmentation is a structured output that contains spatial similarity between source and target domains, adversarial learning is used in the output space rather than the feature space. Furthermore, a multilevel adversarial learning scheme is developed to adapt the features at different levels of the segmentation model.

2) CLAN: Category-level adversaries network (CLAN) [70] exploits category-level adversaries to realize domain adaption on semantic segmentation. Specifically, for samples of already well-aligned categories, the weight of adversarial loss is reduced, thus reducing the side effects of adversarial learning. For samples of categories that are not well-aligned, the weight of adversarial loss is increased.

3) ADVENT: Adversarial entropy (ADVENT) [71] used the idea of adversarial learning to perform domain adaptive semantic segmentation. When the model is only trained on source domain and predicts on images close to the source domain, producing overconfidence is easy, that is, low entropy, and when predicting on images close to the target domain, it is

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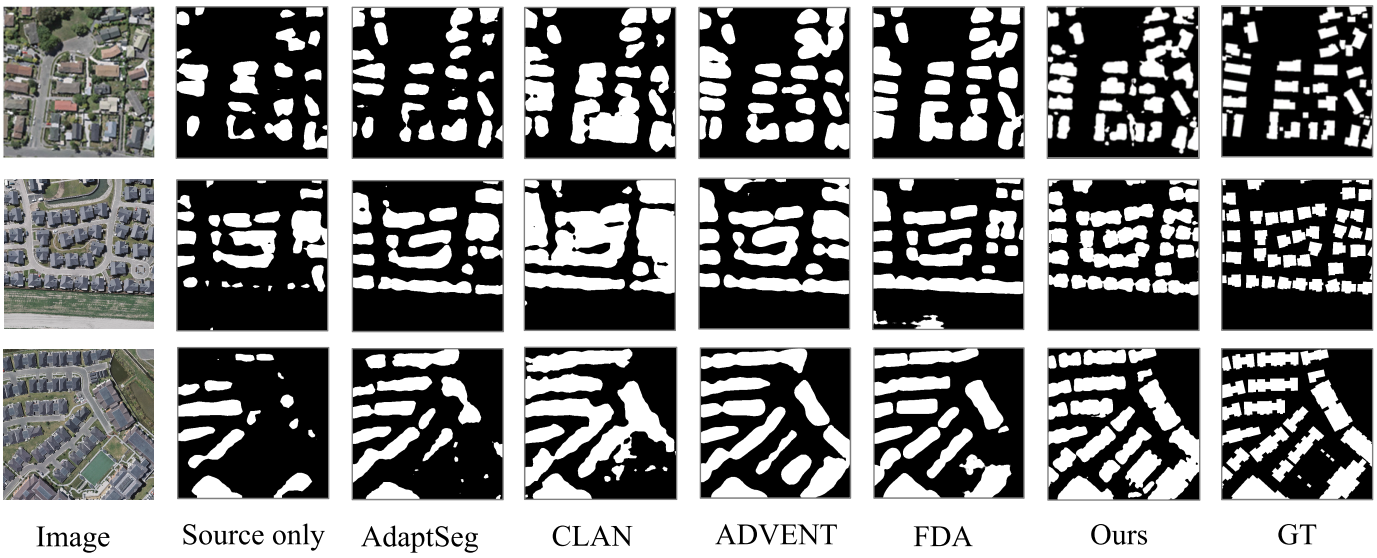

Fig. 4. Qualitative examples for the Urban→WHU case.

often under-confident, that is, high entropy. Thus, the method uses entropy loss to minimize entropy or use adversarial loss directly to minimize entropy indirectly to reduce the gap between the source and target domains.

4) FDANet: FDANet [45] is a full-level UDA method that combines image-, feature-, and output-level information for extracting building in RS images. It uses corresponding strategies at three levels to reduce domain shift to realize knowledge transfer from the source domain to the target domain.

The parameter settings of the compared methods all use the default settings. In addition, the source-only approach, which trains only with source domain data and tests directly on the target domain, uses DeepLabV3+ [38] and ResNet101 [66].

D. Evaluation Metrics

For the quantitative analysis of building extraction results, F1-score (F1) and intersect over union (IoU) are used to evaluate the difference between the ground truth and the predicted maps

$$\mathrm{F1} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{11}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}. \tag{12}$$

E. Result on Different Tasks

1) LoveDA-Urban as Source Dataset: First, to visually compare and analyze, the building extraction results of each method are displayed in Figs. 4 and 5. Through the analysis of the datasets, a large number of modern single-family low-rise buildings are found in the WHU building dataset whereas large-scale high-rise buildings are found in the LoveDA-Urban with obvious offset phenomenon (nonorthophoto). In addition, due to different imaging conditions, the tones of the two datasets are also inconsistent. Obviously, in the presence of such domain gap, when only the source domain data, Urban, is used to train and extract buildings from WHU as the target domain, a large number of missed detections will be noticed, as shown in the second column of Fig. 4. In this case, the problem of image tone difference caused by different imaging conditions can be solved to a certain extent by introducing a method of changing the image style. On the contrary, by aligning the target and source domain features in the feature space, the key knowledge of buildings can be transferred to extract buildings accurately. As shown in columns 3–6, all of these methods utilize feature-level or output-level alignment, and building misses and false detections are reduced. Among them, FDA uses additional image-level alignment, so better results are obtained. However, our method achieves the best results (in column 7) by combining image-level and feature-level strategies from a memory perspective rather than by simply stacking them. Similar to the Urban dataset, the Rural dataset is also sampled from some cities in China, but it only includes rural areas, which means that the image contains a large number of natural objects, relatively few building objects, and single building shapes. Thus, from Urban to Rural, the model is easier to transfer. When only the source domain data are used for training, the buildings can be mainly recognized when testing on the target domain, but the building objects are not completely extracted, as shown in the second column of Fig. 5. Under such tasks, almost all methods can achieve good results (columns 3–7).

As shown in Table II, F1 and IoU are used to analyze the building extraction results for each method quantitatively when Urban is the source data. Consistent with the visualized results, our method achieves the best quantitative performance among other SOTA methods. Overall, all domain adaptation methods achieve significant accuracy improvements. Specifically, AdaptSeg aligns the data distributions of the source and target domains in the feature and output spaces.

Compared with source only, in Urban→Rural case, AdaptSeg achieves an F1 increase of 12.1%, an IoU increase of 11.28%, and suboptimal results. However, on a difficult task, Urban→WHU, it can also be improved but is less effective

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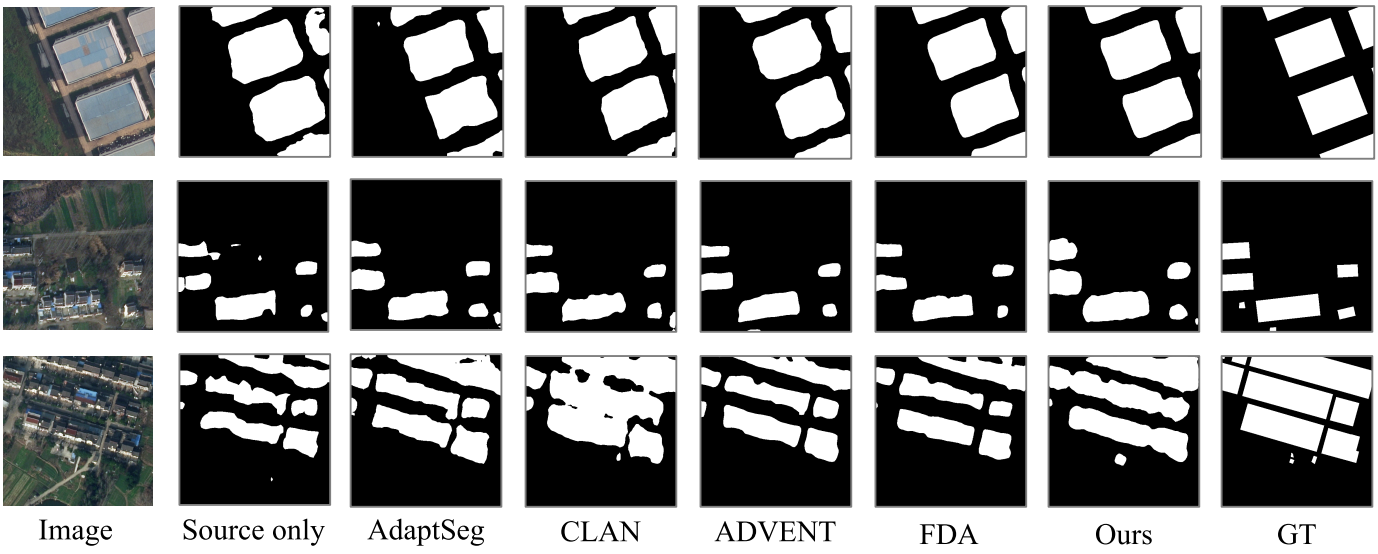

Fig. 5. Qualitative examples for the Urban→Rural case.

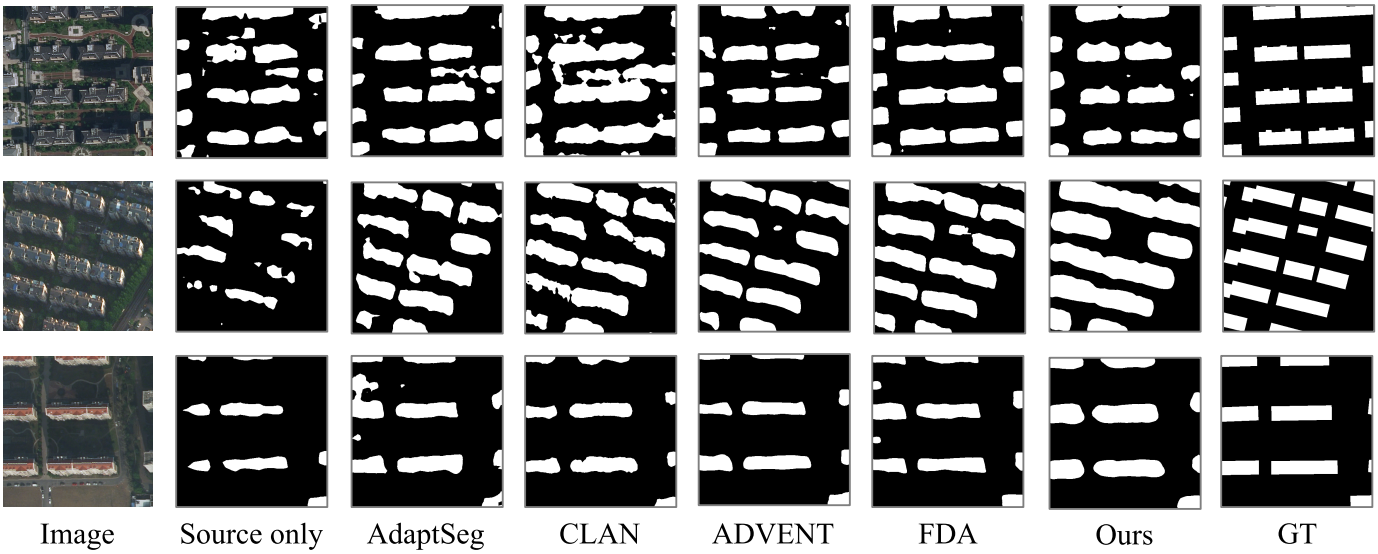

Fig. 6. Qualitative examples for the Rural→Urban case.

TABLE II
QUANTITATIVE ACCURACY WHEN URBAN AS SOURCE DOMAIN

Method	Urban→Rural		Urban→WHU	
	F1	IoU	F1	IoU
Source only	0.6207	0.4541	0.5841	0.4216
AdaptSeg	<u>0.7417</u>	<u>0.5669</u>	0.6633	0.5001
CLAN	0.7041	0.5298	0.6502	0.4909
ADVENT	0.7245	0.5497	0.6772	0.5155
FDA	0.7351	0.5598	<u>0.7263</u>	<u>0.5503</u>
MCD(Ours)	**0.7813**	**0.6171**	**0.7671**	**0.6048**

than the other methods, which obtain an F1 increase of 7.92% and an IoU increase of 7.85%. CLAN is also a feature- and output-level aligning method. Compared with source only, in Urban→Rural tasks, F1 increased by 8.34%, and IoU increased by 7.57%, whereas in Urban→WHU task, F1 increased by 6.61%, and IoU increased by 6.93%. ADVENT is an output-level alignment method that acts on the entropy map of the output. Compared with source only, in Urban→Rural tasks, F1 increased by 10.38%, and IoU increased by 9.56%, whereas in Urban→WHU task, F1 increased by 9.31%, and IoU increased by 9.39%. FDA combines image-, feature-, and output-level alignment to extract building in RS images. Thus, in the difficult case, Urban→WHU, it achieves an F1 increase of 14.28%, an IoU increase of 12.87%, and suboptimal results. In Urban→Rural tasks, the improvement is 11.44% and 10.57%.

Our method, MCD, reduces the gap between the source and target domains from the memory point of view and achieves the best results among all the compared methods with a large F1 increase of 18.3% and a large IoU increase of 18.32% on the Urban→WHU case. The gain is 16.06% in F1 and 16.3% in IoU on the Urban→Rural case. By memorizing the key features of the categories in the source domain and comparing these features stored in the memory bank with the features extracted on the target

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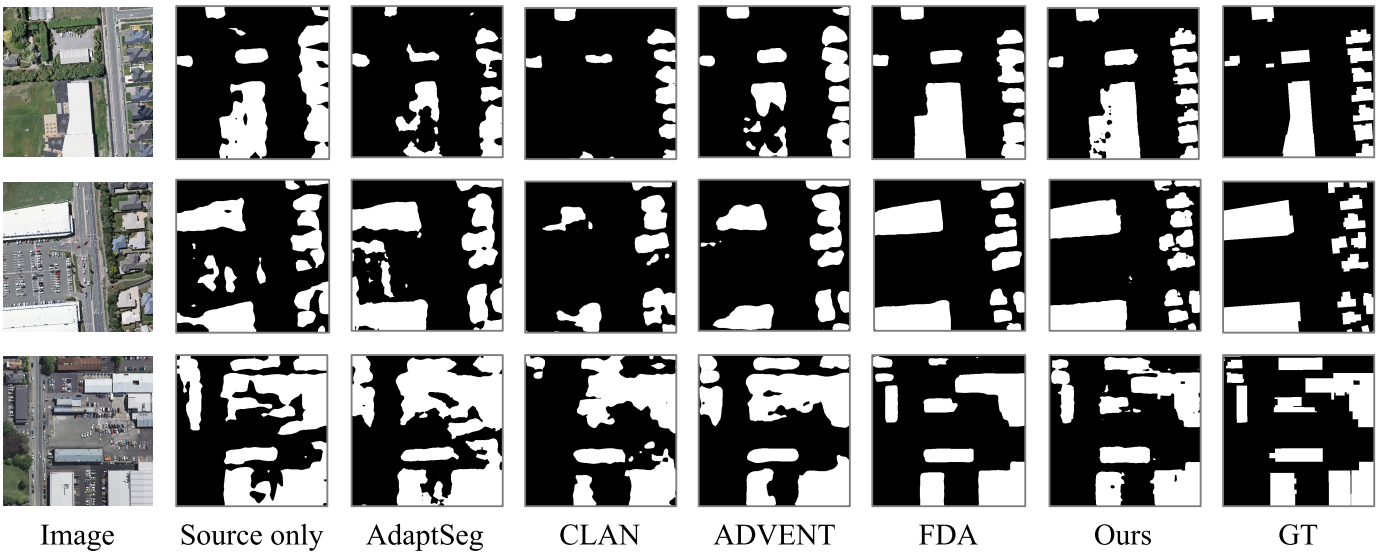

Fig. 7. Qualitative examples for the Rural→WHU case.

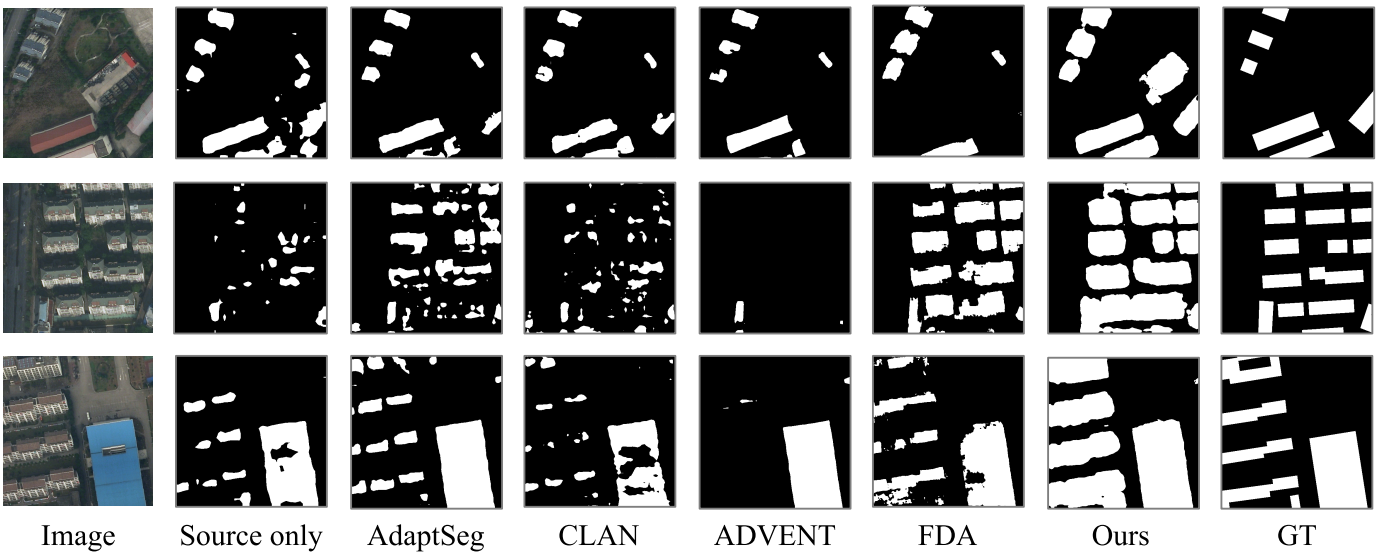

Fig. 8. Qualitative examples for the WHU→Urban case.

domain, the semantic gap can be reduced to achieve knowledge transfer.

2) LoveDA-Rural as Source Dataset: First, to compare multiple methods visually, some typical examples are selected as shown in Figs. 6 and 7. In the Rural→Urban task, by analyzing the images in the dataset, it is not difficult to find that although the sampling areas of the two datasets are all Chinese cities, the domain gap between them is still very obvious. In the Rural dataset, the density of man-made objects is low, the distribution is scattered, and the scale of building objects is relatively small. Hence, as shown in the second column of Fig. 6, when only using the Rural dataset for training and testing on Urban, a lot of misclassification and incomplete identification of buildings will be encountered. By using different domain adaptation methods, as shown in columns 3–6 of Fig. 6, this poor building recognition result is also improved to varying degrees, which can also be confirmed from the quantitative accuracy results (Table III). Nevertheless, these methods are often not stable enough, as shown in the first and third rows of Fig. 6, and the recognition performance may be good in some images, but not satisfactory in others. Although the proposed method also has some instances of misclassification and missing classification, it is more stable and robust overall, so it shows the best accuracy in quantitative results. In the case of Rural→WHU, the sampling area of the WHU in European countries comprises buildings that are often single-family, low-density, and single-style, a shared commonality with the Rural dataset. However, a domain shift between the two datasets is still observed due to differences in sensors and sampling time. By observing the 2–7 columns of Fig. 7, compared with the source-only method, the strategy of using domain adaptation can achieve better results, whereas our method greatly reduces the missed detection and false detection rate and can extract smoother building boundaries. As reported in Table III, our method obtains the best performance against other SOTA methods.

3) WHU as Source Dataset: Figs. 8 and 9 show the quantitative results when the WHU dataset is tested on the two other datasets as the source domain. As previously analyzed, the buildings in the WHU dataset are low, low-density,

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Fig. 9. Qualitative examples for the WHU→Rural case.

TABLE III
QUANTITATIVE ACCURACY WHEN RURAL AS SOURCE DOMAIN

Method	Rural→Urban		Rural→WHU	
	F1	IoU	F1	IoU
Source only	0.5091	0.3410	0.6013	0.4562
AdaptSeg	0.6105	0.4436	0.6145	0.4632
CLAN	0.5796	0.4178	0.6637	0.5011
ADVENT	0.6508	0.4911	0.7329	0.5581
FDA	0.6706	0.5045	0.7787	0.6376
MCD(Ours)	**0.7343**	**0.5601**	**0.8006**	**0.6853**

TABLE IV
QUANTITATIVE ACCURACY WHEN WHU AS SOURCE DOMAIN

Method	WHU→Urban		WHU→Rural	
	F1	IoU	F1	IoU
Source only	0.3392	0.2009	0.4717	0.3104
AdaptSeg	0.4627	0.3146	0.4898	0.3243
CLAN	0.4695	0.3086	0.5058	0.3376
ADVENT	0.4511	0.2954	0.4731	0.3129
FDA	0.6290	0.4778	0.5102	0.3424
MCD(Ours)	**0.7277**	**0.5507**	**0.7205**	**0.5465**

and relatively single in style, whereas the buildings in the Urban dataset are just the opposite. Therefore, only very poor building recognition results were obtained when training with source-only domain data, both in terms of quantitative results and precision. In the WHU→Urban task, some domain adaptation methods (columns 3–5 of Fig. 8) cannot also achieve satisfactory results, and the number of missed and false detections is large. This finding shows that these methods fail to mine information effectively to facilitate the transfer of building knowledge from the source domain to the target domain. Only the FDA and our method can achieve relatively impressive results. However, the buildings extracted by FDA often have some holes and incompleteness, whereas our method do not have. With WHU as the source domain and Rural as the target domain (Table IV), except our method, the accuracy improvement of all methods is very limited. To sum up, our method can memorize the building features of the source domain through a series of constraints and can mine the essential features of the building to realize the knowledge transfer on the target domain.

All quantitative results are presented in Table V, where the best results are shown in bold and the next best are underlined. Our method achieves the best results in all cases of the experiment. FDA obtained the second-best results, and the three other DA methods have limited improvement. From the qualitative results in Figs. 4–9, it can be found that the buildings extracted by our method are basically complete, and the missed and false detections are relatively few, which shows that our method can successfully implement transfer from the source domain to the target domain. Although other methods have improved accuracy compared with source only, the extracted buildings all have missed detections, false detections, incomplete object extraction, and unclear boundaries to a certain extent, which means that these methods are not yet stable and robust enough.

To evaluate the model from the perspective of efficiency, the parameters and floating-point operations per second (FLOPs) of each method are counted (Table VI). AdaptSegNet and ADVENT have the largest number of parameters, and the FLOPs are not low, but they cannot achieve stable building extraction results in all experimental situations. The FLOPs of CLAN are higher, but some cases have the above two method types. FDA's parameters and FLOPs are the lowest, and it can achieve better results in most cases, but achieving satisfactory results is difficult when WHU is used as the source domain. The number of parameters of our method is relatively low because the MCD module we designed does not need parameters for learning, and the main sources of parameters are the discriminator and the teacher–student network. Although the FLOPs are high and the computational load is large, the best results can be achieved in various situations, especially when

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TABLE V
SUMMARY OF RESULT EVALUATION

Method	Urban→Rural		Urban→WHU		WHU→Rural		WHU→Urban		Rural→Urban		Rural→WHU	
	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU
Source only	0.6207	0.4541	0.5841	0.4216	0.3392	0.2009	0.4717	0.3104	0.5091	0.3410	0.6013	0.4562
AdaptSeg	0.7417	0.5669	0.6633	0.5001	0.4627	0.3146	0.4898	0.3243	0.6105	0.4436	0.6145	0.4632
CLAN	0.7041	0.5298	0.6502	0.4909	0.4695	0.3086	0.5058	0.3376	0.5796	0.4178	0.6637	0.5011
ADVENT	0.7245	0.5497	0.6772	0.5155	0.4511	0.2954	0.4731	0.3129	0.6508	0.4911	0.7329	0.5581
FDA	0.7351	0.5598	0.7263	0.5503	0.6290	0.4778	0.5102	0.3424	0.6706	0.5045	0.7787	0.6376
MCD(Ours)	**0.7813**	**0.6171**	**0.7671**	**0.6048**	**0.7277**	**0.5507**	**0.7205**	**0.5465**	**0.7343**	**0.5601**	**0.8006**	**0.6853**

TABLE VI
MODEL PARAMETERS AND TRAINING EFFICIENCY

Method	AdaptSeg	CLAN	ADVENT	FDA	Ours
Parameters(M)	186	166	186	79	108
FLOPs (GMac)	186	193	186	137	201

WHU is used as the source domain, the IoU can be more than 20% higher than the suboptimal FDA. Therefore, we have reason to believe that it is worth it.

Fig. 10 shows that our method can also correct some errors in labels. In the red boxed area in the first row, the impervious surface is labeled incorrectly as a building, but it is not recognized as a building in the recognition results of our method. The second row shows the omissions in the labels, and our method can also identify unlabeled buildings in the labels. This finding proves that our method can mine the representative features of buildings.

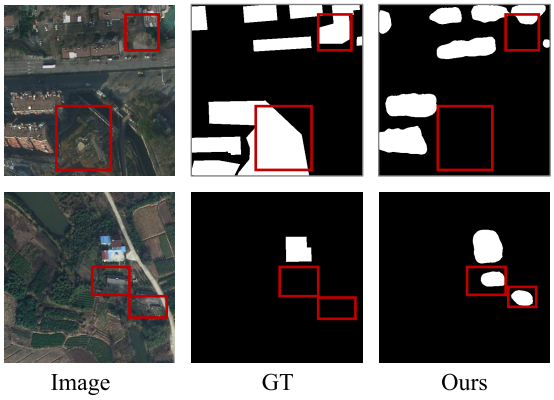

Fig. 10. Correct some errors.

F. Discussion

1) Ablation Study: To explore how the various modules used and designed can benefit the set cross-domain building extraction task, ablation experiments are conducted. Table VII shows the results on the Urban→Rural task. Only the teacher–student model and the generated pseudo-labels are used when none of these three modules are exploited. Compared with the source-only method, the accuracy improvement is very small, where F1 and IoU are improved by 0.82% and 0.67%, respectively. The reason is that most of what the student network learns is the knowledge of the source domain, which makes it difficult to adapt to the target domain data. Naturally, the generated pseudo-labels of the target domain are inaccurate, and it is difficult to play a correct supervision role. When only WF is used to transform the input source domain image, the accuracy is improved slightly by 2.48% and 2.79% on F1 and IoU because the distribution of the source domain can be pulled closer to the distribution of the target domain to a certain extent through the WF operation. On this basis, the discriminator is used at the feature level to narrow the domain gap between the source domain and target domain features, which brings a significant improvement in accuracy, with F1 and IoU improved by 5.87% and 6.60%, respectively.

When only MCD is used, the accuracy improvement is very obvious, in which F1 and IoU are improved by nearly 10% compared with the source-only method. By getting closer to the source and target domain features of memory, a compromise can be found to achieve partial knowledge transfer. However, due to the lack of feature-level constraints, the model's perception and extraction of domain-invariant features are not sufficient, which is not advantageous compared with other DA methods. Obviously, on this basis, the use of the discriminator to perform a constraint before the feature is inputted into the MCD can ensure the memory of the domain-invariant features. Thus, the accuracy is improved again and surpasses SOTA. When all modules are used together, the image-level and feature-level constraints work together to assist MCD in the storage of domain-invariant features extracted from the source domain and their comparison with the target domain features.

TABLE VII
RESULTS OF THE ABLATION STUDIES ON THE URBAN→RURAL TASK

WF	D_f	MCD	F1	IoU
			0.6289	0.4608
√			0.6455	0.4820
√	√		0.6794	0.5201
		√	0.7197	0.5452
	√	√	0.7638	0.5867
√	√	√	0.7813	0.6171

In addition, to demonstrate the effectiveness of the memory bank, we compared the quantitative results of full model with the model without memory bank on six tasks. Tables VIII–XII show the results of ablation studies on these tasks. Only relying

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TABLE VIII
RESULTS OF THE ABLATION STUDIES ON THE URBAN→WHU TASK

WF	D_f	MCD	F1	IoU
			0.5991	0.4325
√			0.6326	0.4592
√	√		0.6702	0.5119
		√	0.7200	0.5428
	√	√	0.7514	0.5806
√	√	√	0.7671	0.6048

TABLE IX
RESULTS OF THE ABLATION STUDIES ON THE WHU→RURAL TASK

WF	D_f	MCD	F1	IoU
			0.3411	0.2213
√			0.3753	0.2586
√	√		0.4643	0.3002
		√	0.6207	0.4629
	√	√	0.6635	0.5100
√	√	√	0.7277	0.5507

TABLE X
RESULTS OF THE ABLATION STUDIES ON THE WHU→URBAN TASK

WF	D_f	MCD	F1	IoU
			0.4748	0.3184
√			0.4836	0.3229
√	√		0.4899	0.3257
		√	0.6693	0.5021
	√	√	0.7118	0.5308
√	√	√	0.7205	0.5465

TABLE XI
RESULTS OF THE ABLATION STUDIES ON THE RURAL→URBAN TASK

WF	D_f	MCD	F1	IoU
			0.5088	0.3402
√			0.5263	0.3499
√	√		0.5607	0.4053
		√	0.6527	0.5000
	√	√	0.7039	0.5312
√	√	√	0.7343	0.5601

on the teacher–student model cannot guarantee the transferability of the extracted knowledge. For example, on the Urban→Rural task, the features extracted from source domain can meet requirements of contrastive learning without memory bank, but only when the current batch of data provides self-supervised information. Therefore, the accuracy is reduced compared to the full model with memory bank. On the Rural→Urban task, since the number of buildings in the rural

TABLE XII
RESULTS OF THE ABLATION STUDIES ON THE RURAL→WHU TASK

WF	D_f	MCD	F1	IoU
			0.5898	0.4419
√			0.6379	0.4817
√	√		0.6788	0.5210
		√	0.7305	0.5478
	√	√	0.7693	0.5867
√	√	√	0.8006	0.6853

TABLE XIII
EFFICIENCY OF MEMORY BANK

IoU	U-R	U-W	W-U	W-R	R-U	R-W
M.B	0.6171	0.6048	0.5465	0.5507	0.5601	0.6853
w/o M.B	0.5947	0.5684	0.4916	0.4382	0.4968	0.6306

TABLE XIV
RESULTS ON DIFFERENT CONFIDENCE VALUES OF CONSTRUCTING MEMORY BANK

C.v(M. B)	0.8	**0.85**	0.9
F1	0.7801	**0.7813**	0.7742
IoU	0.6145	**0.6171**	0.6100

dataset is relatively small, the number of building features obtained from the source domain is inevitably reduced, resulting in a loss of accuracy. However, by using the features stored in memory bank to conduct contrastive learning with the features extracted by student network, the student network can be promoted to extract more correct knowledge. Subsequently, the teacher network can be updated with the assistance of student network and becomes stronger and stronger via continuous information interaction, which eventually makes the features stored in memory bank to become more and more stable and robust. Table XIII shows that the memory bank is effective on all tasks, thus proving the modules we designed are feasible and efficient.

2) Parameter Analysis: For the confidence value of constructing memory bank (Section III-D), different values are set for experiments. We conduct related experiments on the Urban→Rural task. Usually, the higher the confidence level is, the more reliable the result will be, and the less likely it is to be misjudged. Our aim is to store domain-invariant building features in a memory bank. If the confidence level is increased blindly, although the correctness of building features can be ensured, the diversity of building features is ignored. Some difficult samples may be filtered out, making building features more biased toward these high-confidence samples, thus affecting the expression of domain-invariant features. As shown in Table XIV, when the confidence threshold is set to 0.8, the memory bank is very likely to contain misclassified features, thus reducing the accuracy of the entire model. When the

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TABLE XV
RESULTS ON DIFFERENT CAPACITIES OF MEMORY BANK

Size	64	128	256	512
F1	0.7624	0.7709	0.7813	**0.7852**
IoU	0.5821	0.5984	0.6171	**0.6190**

threshold is set to 0.9, since the stored domain-invariant features are biased at this time, the model accuracy also drops. Thus, through experiments, 0.85 was chosen as the threshold for selecting the features stored in the memory bank.

Further, the capacity of the memory bank is also explored. A larger capacity means that more feature vectors will be selected for the comparison loss calculation. On the one hand, the calculation of the loss will be more comprehensive, and the accuracy will be improved. On the other hand, the amount of calculation will also increase. Therefore, to improve efficiency, it is necessary to find a balance between accuracy and calculation amount. As shown in Table XV, as the capacity increases, the accuracy also improves, specifically, from 64 to 128, and F1 increases by 0.85%. When the size of the memory bank was increased from 128 to 256, F1 improved by 1.04%. From 256 to 512, the change in accuracy is significantly smaller, and F1 only rose by 0.39%. Therefore, 256 is used as the capacity size of the memory bank.

V. CONCLUSION

In this study, to improve the generalization ability of extracting buildings from HRSIs in different domains, a novel and effective MDANet, which is inspired by memory mechanisms, is designed to achieve this purpose. Our method first uses a simple image-level alignment method to transfer the image style, initially reducing the gap between the source domain and the target domain and making it easier to gain the correlative memory. Then, a discriminator is designed to align the source and domain data distributions to extract more transferable memory features. Subsequently, the MCD module is designed to reuse and update the memory stored in the memory bank to realize the transfer of the key knowledge of the building.

Our experiments demonstrated that exploiting the memory mechanism can have a very significant positive effect on the task. In the future work, we will continue to explore the relationship between memory mechanism and domain adaptation tasks and attempt to use more efficient strategies to optimize the domain adaptation model.

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